Gene Length Bias in Human Papillomavirus Integration Sites: A Statistical Analysis Using KNIME

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Abstract

Integration of high-risk Human Papillomavirus (HPV) into the human genome represents a critical event in carcinogenesis. While recurrent integration hotspots are well documented, the structural determinants governing host gene susceptibility remain incompletely defined. Here, we evaluate whether human gene length influences HPV integration probability using VISDB integration datasets for HPV16 and HPV18 merged with Ensembl BioMart annotations. Statistical analysis performed in KNIME reveals that HPV-integrated genes exhibit significantly greater length than non-integrated genes (Wilcoxon–Mann–Whitney, $p = 0.0001$). Distributional analysis demonstrates a consistent rightward shift in log-transformed gene length among integrated genes. Genomic validation using the UCSC Genome Browser confirms integration within established oncogenic regions, including 8q24 near *MYC*. These findings identify gene length as a measurable structural bias contributing to HPV integration susceptibility, while underscoring the role of genomic context in hotspot formation.

Keywords: HPV integration, gene length bias, viral oncogenesis, genomic hotspots, statistical genomics, HPV16, HPV18

1 Introduction

High-risk HPV types, particularly HPV16 and HPV18, promote malignant transformation through integration into host genomic DNA. Integration events can disrupt tumor suppressor genes, amplify oncogenic loci, and alter regulatory architecture. Recurrent hotspots such as *PVT1*, *FHIT*, *MACROD2*, and loci within 8q24 have been consistently reported. However, whether intrinsic structural properties of host genes influence integration frequency remains unclear.

We hypothesized that longer genes, by occupying larger genomic intervals, present increased opportunity for viral integration.

2 Materials and Methods

2.1 Data Acquisition

HPV16 and HPV18 integration site data were obtained from VISDB. Human gene annotations were retrieved from Ensembl BioMart (GRCh38). Integration coordinates were mapped to corresponding host genes using genomic overlap criteria.

2.2 Computational Workflow

All preprocessing and statistical analyses were performed using the KNIME Analytics Platform. Genes were categorized as:

- Integrated genes
- Non-integrated genes

Gene length was calculated from genomic start and end positions and log-transformed to correct for skewed distribution.

2.3 Statistical Analysis

Gene length distributions between integrated and non-integrated groups were compared using the Wilcoxon–Mann–Whitney test. Statistical significance was defined at $p < 0.05$.

3 Results

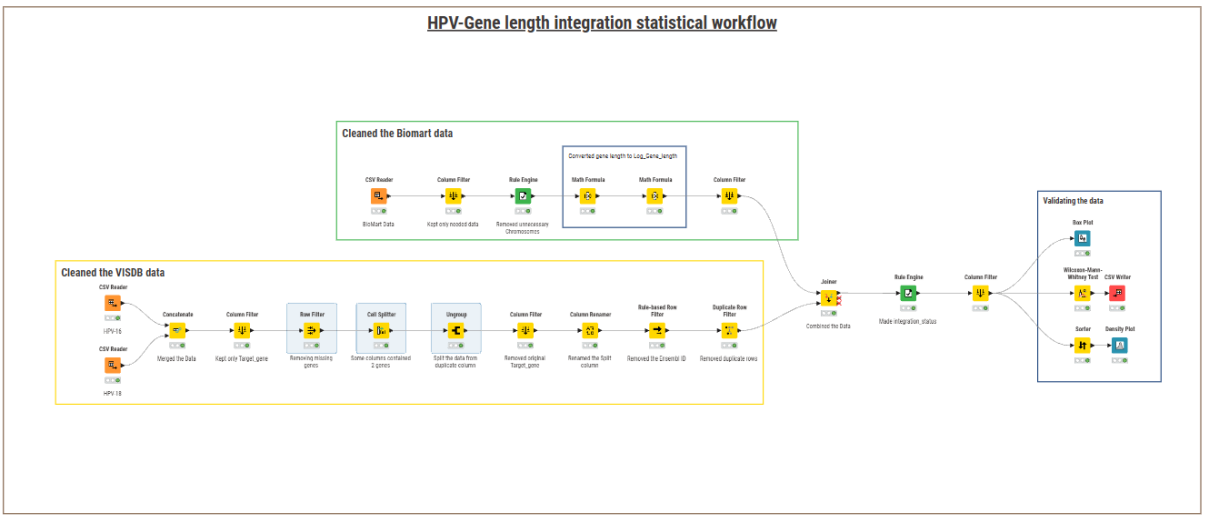
3.1 Gene Length Distribution Bias

Integrated genes demonstrated higher median log-transformed lengths relative to non-integrated genes.

Statistical testing confirmed a significant difference between groups ($/alpha = 0.05$), supporting a gene length bias in HPV integration.

3.2 Hotspot Context

Genomic inspection using the UCSC Genome Browser revealed that several integration hotspots reside within established oncogenic regions, including 8q24 proximal to *MYC*. Importantly, certain shorter genes within highly active regulatory domains also exhibited integration events, suggesting that chromatin state and regulatory architecture influence integration beyond structural length bias.



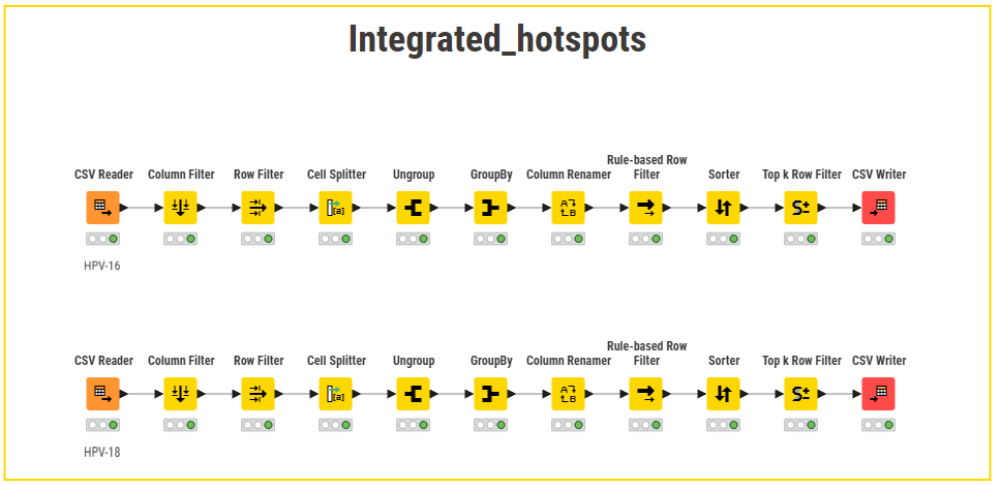


Figure 1: KNIME workflow illustrating data integration, gene classification, log transformation, and Wilcoxon–Mann–Whitney statistical testing. (This workflow is available on GitHub link:- https://github.com/iamsahill4/HPV-KNIME-Workflow)

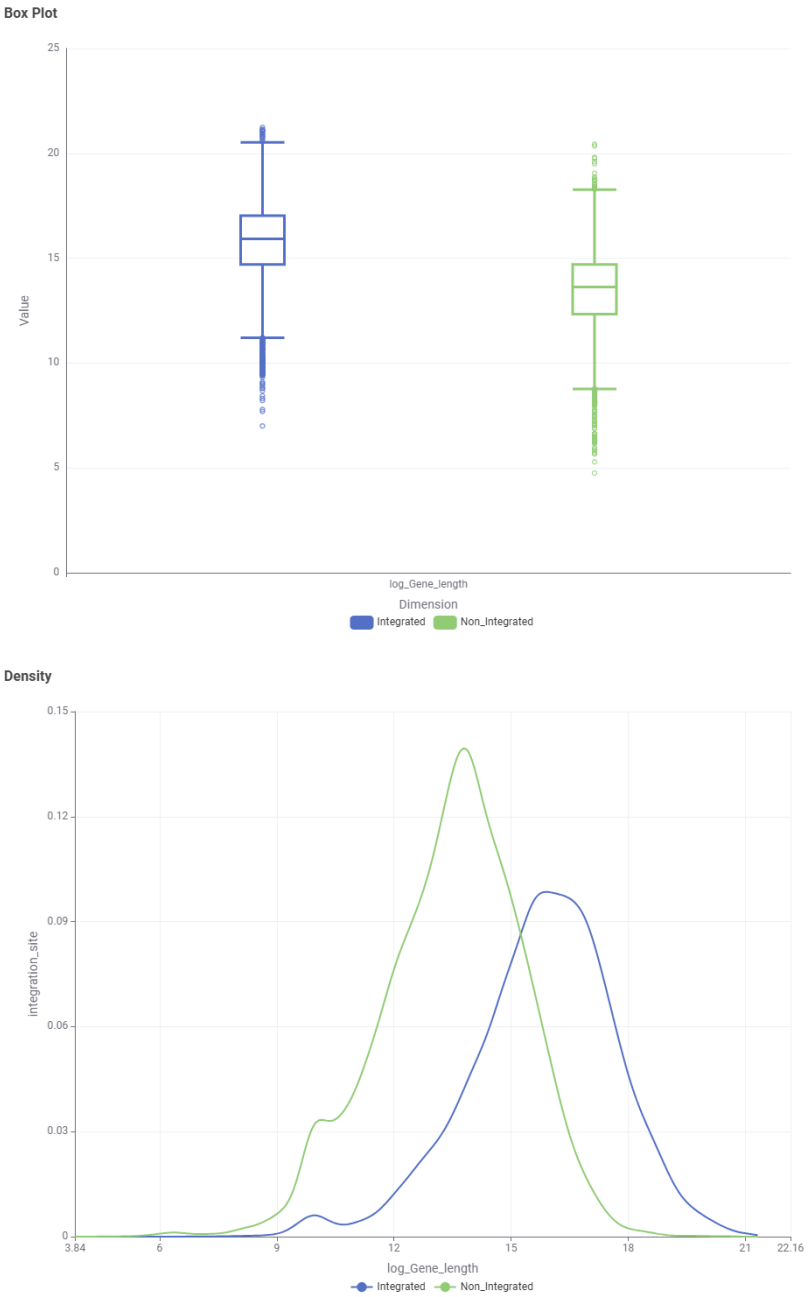


Figure 2: Distributional analysis of log-transformed gene length. Top: Box plot comparing integrated and non-integrated genes. Bottom: Kernel density estimation demonstrating rightward shift among integrated genes.

4 Discussion

Our findings demonstrate that human gene length is significantly associated with HPV integration susceptibility. The most direct interpretation is probabilistic: longer genes span greater genomic territory and therefore present increased opportunity for viral insertion events.

However, gene length alone does not fully explain hotspot localization. Integration within shorter genes positioned in transcriptionally active or enhancer-rich regions indicates that epigenetic and chromatin features likely refine integration targeting. These observations support a composite model in which structural exposure establishes baseline susceptibility, while regulatory context determines hotspot specificity.

5 Conclusion

Human gene length is positively associated with HPV integration probability. Integrated genes are significantly longer than non-integrated genes. Although gene length contributes to susceptibility, integration hotspot formation likely arises from interaction between structural and regulatory genomic features.

6 Limitations

This analysis evaluated a single genomic feature (gene length) and did not incorporate multivariate modeling or experimental validation.

7 Future Directions

Future investigations should integrate chromatin accessibility, histone modification profiles, gene expression data, replication timing, and three-dimensional genome organization to develop predictive models of HPV integration susceptibility.

Conflict of Interest

The authors declare no competing interests.

Data Availability

All datasets used in this study are publicly available from VISDB, Ensembl BioMart, and UCSC Genome Browser.